(1) $n_1 = 1.8 \quad n_2 = 1.0 \quad \theta_i = 30^\circ \quad (TM)$

$$\theta_t = \sin^{-1}\left(\frac{n_1}{n_2}\sin\theta_i\right) = 64.2^\circ$$

$$T = 1 - R = 1 - |\Gamma_{TM}|^2 = 1 - \left(\frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}\right)^2 = 0.9975$$

For total reflection, $\theta_i \geq \theta_c = \sin^{-1}(n_2/n_1) = 33.75^\circ$

(2) At 1.8 eV ($\lambda = 688.8$ nm)

$n' = 3.785 \quad n'' = 0.151 \Rightarrow n = 3.785 - j\,0.151$

a) $v = c/n' = 0.792 \times 10^8$ m/s

$$\alpha = \frac{\omega}{c} n'' = \frac{2\pi \times 4.35 \times 10^{14}}{3 \times 10^8}(0.151) = 1.38 \times 10^6 \text{ 1/m}$$

b) $0.01 = e^{-2\alpha z} \Rightarrow z = \frac{\ln(0.01)}{-2\alpha} = 1.67 \times 10^{-6}$ m

(3) $d = 100\ \mu m \quad L = 5 \quad$ Blue $\lambda_b \quad$ red λ_r

$$x_{1_b} = L\lambda_b/d \qquad x_{1_r} = L\lambda_r/d$$

$$x_{1r} - x_{1b} = \frac{L}{d}(\lambda_r - \lambda_b) \approx \frac{5}{100\mu}(0.68 - 0.47) \approx 1.1 \text{ cm}$$

(4) $I_0 = 800$ W/m^2 ; $I_1 = \frac{800}{2} = 400$ W/m^2 ; $I_2 = 400\cos^2(30) = 300$ W/m^2

$I_3 = 300\cos^2(90) = 0$

(5) $f = 45/2 = +22.5$ cm $\quad d_i = -35$ cm

$$\frac{1}{d_o} = \frac{1}{d_i} + \frac{1}{f} = \frac{1}{35} + \frac{1}{22.5} \Rightarrow d_o = 13.7 \text{ cm}$$

(6) Photon energy $= \hbar\omega = 2.755$ eV $\Rightarrow$ KE $= \hbar\omega - W = 0.355$ (eV)

Electron flux = Photon flux $= \frac{0.5\text{W}}{2.755\text{eV}}$

$I = 0.181$ A